

Notes: Mankiw and Reis (QJE, 2002)

Sticky Information versus Sticky Prices: A Proposal to Replace the New Keynesian Phillips Curve

Contents

1 Big picture

2 Data and measurement (key objects)

- 2.1 Why this is a theory paper but still has “measured” objects
- 2.2 The key primitives
- 2.3 The firm’s desired price
- 2.4 The benchmark sticky-price environment
- 2.5 The sticky-information environment
- 2.6 Aggregate demand closure and benchmark comparison models
- 2.7 Calibration and empirical benchmarks
- 2.8 What the paper treats as the core stylized facts

3 Models and empirical specifications (all equations + interpretation)

- 3.1 Economic mechanism and hypotheses
- 3.2 The standard sticky-price model and the New Keynesian Phillips curve
- 3.3 The sticky-information model and the sticky-information Phillips curve
- 3.4 Why strategic complementarity matters
- 3.5 The McCallum critique and the natural-rate property
- 3.6 Experiment 1: a permanent drop in the level of aggregate demand
- 3.7 Experiment 2: an unanticipated disinflation
- 3.8 Experiment 3: an anticipated and credible disinflation
- 3.9 Realistic monetary shocks
- 3.10 Inflation persistence
- 3.11 The acceleration phenomenon

4 Identification and instruments (what makes the estimates credible)

- 4.1 There is no external instrument
- 4.2 What identifies the baseline comparison
- 4.3 Why the sticky-information model looks more credible than the sticky-price model
- 4.4 Evidence on regime changes and credibility
- 4.5 Evidence on slow dissemination of information
- 4.6 Microfoundations and bounded rationality
- 4.7 What the paper can and cannot distinguish
- 4.8 Relation to the literature

5 Tables and figures

- 5.1 Figure I: a permanent fall in the level of aggregate demand
- 5.2 Figure II: an unanticipated disinflation
- 5.3 Figure III: an anticipated disinflation
- 5.4 Figure IV: impulse responses to a realistic monetary shock
- 5.5 Table I: inflation autocorrelations
- 5.6 Table II: the acceleration phenomenon
- 5.7 Table III: sensitivity analysis for the acceleration result

6 Short summary

7 Conclusion

- 7.1 Core conclusion (what the paper establishes)
- 7.2 Economic intuition
- 7.3 Takeaway

1 Big picture

- **Question.** Can a model in which firms update information only intermittently explain inflation-output dynamics better than the standard sticky-price New Keynesian Phillips curve?
- **Setting.** This is a macro theory paper. The core comparison is between three models: the standard sticky-price model, a new sticky-information model, and a backward-looking Phillips curve benchmark. The authors study their implications for disinflation, monetary shocks, inflation persistence, and the “acceleration phenomenon.”
- **Core challenge.** The standard sticky-price model has a hard time matching three widely accepted facts: credible disinflations are usually recessionary rather than expansionary, monetary policy affects inflation with a delay, and inflation tends to rise when economic activity is strong.
- **Main conceptual contribution.** The paper shifts the source of nominal rigidity from *prices* to *information*. Prices can change every period, but many firms keep following plans based on outdated information.
- **Modeling idea.** Each period, a fraction λ of firms updates its information set and recomputes optimal prices; the remaining firms continue to price using earlier information and old plans.
- **Main theoretical result.** The sticky-information Phillips curve depends on current output, expected inflation, and expected output growth, but the key expectations are *past expectations of current conditions*, not current expectations of future conditions.
- **Main policy result 1.** Unlike the sticky-price model, the sticky-information model implies that disinflations are always contractionary, although anticipated and credible disinflations are less costly than surprise disinflations.
- **Main policy result 2.** Under a realistic stochastic process for money growth, monetary shocks have their largest effect on inflation only after a substantial delay — about seven quarters in the baseline calibration.
- **Main empirical-style result.** The sticky-information model reproduces the positive correlation between output and the *change* in inflation that macroeconomists often emphasize in reduced-form Phillips-curve evidence; the sticky-price model predicts the wrong sign.
- **Main credibility result.** The authors show that the model lines up with conventional central-bank views and with suggestive evidence on slow adjustment of expectations, information frictions, and credibility effects.
- **Economic takeaway.** Macroeconomic inertia may come less from firms being unable to change prices and more from firms being slow to absorb, interpret, and act on new information.

2 Data and measurement (key objects)

2.1 Why this is a theory paper but still has “measured” objects

- This is not an empirical panel paper. There is no regression sample, no external instrument, and no treatment-control design.

- The paper instead studies a set of **model objects** and compares their implications to well-known macro facts.
- The relevant “measured” objects are therefore:
 - the New Keynesian Phillips curve and the sticky-information Phillips curve,
 - impulse responses of output and inflation to different monetary experiments,
 - autocorrelations of inflation,
 - and correlations between output and changes in inflation in U.S. data.

Interpretation. The paper is asking which nominal-rigidity mechanism generates empirically plausible dynamics, not estimating a reduced-form causal effect from cross-sectional variation.

2.2 The key primitives

- p_t : aggregate price level.
- $\pi_t = p_t - p_{t-1}$: inflation.
- y_t : output gap, with potential output normalized to zero.
- m_t : nominal aggregate demand, interpreted either as money supply or more broadly as nominal GDP.
- α : sensitivity of a firm’s desired relative price to output.
- λ : fraction of firms adjusting prices in the sticky-price model or, in the sticky-information model, fraction of firms updating information each period.

Interpretation. The model uses the same two parameters, α and λ , to make the comparison between sticky prices and sticky information as clean as possible.

2.3 The firm’s desired price

In both models, the firm’s desired log price is

$$p_t^* = p_t + \alpha y_t. \tag{1}$$

This says that a firm’s desired relative price $p_t^* - p_t$ rises in booms and falls in recessions.

A useful equivalent expression, using $m_t = p_t + y_t$, is

$$p_t^* = (1 - \alpha)p_t + \alpha m_t. \tag{2}$$

Interpretation. When α is small, firms care a lot about what other firms are charging. This strategic complementarity is what eventually generates inflation inertia in the sticky-information model.

2.4 The benchmark sticky-price environment

The standard Calvo-style sticky-price model has three objects:

- the desired price p_t^* ,
- the reset price x_t chosen by firms that get to adjust,
- the aggregate price level p_t , which averages across vintages of previously set prices.

The reset price is a discounted average of current and future desired prices,

$$x_t = \lambda \sum_{j=0}^{\infty} (1 - \lambda)^j E_t p_{t+j}^*, \quad (3)$$

and the aggregate price level is

$$p_t = \lambda \sum_{j=0}^{\infty} (1 - \lambda)^j x_{t-j}. \quad (4)$$

Interpretation. Here, firms may be forward-looking and fully informed, but many cannot reset prices each period.

2.5 The sticky-information environment

The alternative model changes one assumption: every firm can post a price each period, but only a fraction λ updates information and recomputes an optimal plan.

If a firm last updated information j periods ago, it sets

$$x_t^j = E_{t-j} p_t^*. \quad (5)$$

The aggregate price level is therefore

$$p_t = \lambda \sum_{j=0}^{\infty} (1 - \lambda)^j x_t^j = \lambda \sum_{j=0}^{\infty} (1 - \lambda)^j E_{t-j} (p_t + \alpha y_t). \quad (6)$$

Interpretation. Prices are not fixed, but many prices are chosen using stale information. This makes the whole price level sluggish even though each firm can still post a price every period.

2.6 Aggregate demand closure and benchmark comparison models

To study dynamics, the authors close the model with a simple quantity-theory style aggregate-demand equation,

$$m_t = p_t + y_t. \quad (7)$$

They compare the sticky-information model to two benchmarks:

- the standard sticky-price model,

- a backward-looking Phillips curve,

$$\pi_t = \beta y_t + \pi_{t-1}, \quad (8)$$

where the paper sets

$$\beta = \frac{\alpha \lambda^2}{1 - \lambda}. \quad (9)$$

Interpretation. The backward-looking model is not microfounded in the paper; it is included because it is empirically popular and offers a benchmark with built-in inflation inertia.

2.7 Calibration and empirical benchmarks

For the dynamic experiments, the paper works at quarterly frequency and chooses:

- $\alpha = 0.1$,
- $\lambda = 0.25$,
- so firms update about once per year on average.

For the stochastic money-growth exercise, the paper assumes

$$\Delta m_t = \rho \Delta m_{t-1} + \varepsilon_t, \quad (10)$$

with

- $\rho = 0.5$,
- standard deviation of ε_t equal to 0.007.

These values are motivated by quarterly U.S. data from 1960 to 1999 on M1, M2, and nominal GDP growth.

Interpretation. The calibration is not designed to fit every moment exactly. It is chosen to generate a plausible, realistic benchmark for comparing the three models' dynamic responses.

2.8 What the paper treats as the core stylized facts

The paper repeatedly evaluates models against three macro facts:

- credible disinflations tend to be recessionary,
- inflation reacts to monetary policy with a long lag,
- vigorous economic activity tends to be associated with rising inflation.

It also looks at inflation persistence, but the paper argues that persistence by itself is not the decisive margin, because serial correlation in money growth can generate high inflation autocorrelation in several models.

3 Models and empirical specifications (all equations + interpretation)

3.1 Economic mechanism and hypotheses

The paper distinguishes two rival mechanisms for nominal rigidity:

- **Sticky prices:** firms are forward-looking and informed, but many cannot change prices.
- **Sticky information:** firms can always post a price, but many update their information and plans only infrequently.

The sticky-information mechanism is motivated by costs of acquiring information, reoptimizing, or simply “thinking” about what new macro information implies for optimal pricing.

The central hypothesis is that informational sluggishness — not just mechanical price adjustment costs — is what produces realistic inflation inertia.

3.2 The standard sticky-price model and the New Keynesian Phillips curve

The three equations are:

$$p_t^* = p_t + \alpha y_t, \tag{11}$$

$$x_t = \lambda \sum_{j=0}^{\infty} (1 - \lambda)^j E_t p_{t+j}^*, \tag{12}$$

$$p_t = \lambda \sum_{j=0}^{\infty} (1 - \lambda)^j x_{t-j}. \tag{13}$$

Solving these yields the standard New Keynesian Phillips curve:

$$\pi_t = \frac{\alpha \lambda^2}{1 - \lambda} y_t + E_t \pi_{t+1}. \tag{14}$$

Interpretation. Inflation today depends on current slack and *current expectations of future inflation*. That forward-looking structure is precisely what generates the model’s counterfactual responses to disinflation and monetary shocks.

3.3 The sticky-information model and the sticky-information Phillips curve

The sticky-information model uses the same desired price equation,

$$p_t^* = p_t + \alpha y_t, \tag{15}$$

but assumes that only a fraction λ of firms updates information each period. Firms that last updated j periods ago choose

$$x_t^j = E_{t-j} p_t^*. \quad (16)$$

Aggregating over all information vintages gives

$$p_t = \lambda \sum_{j=0}^{\infty} (1 - \lambda)^j E_{t-j} (p_t + \alpha y_t). \quad (17)$$

After algebra, the implied sticky-information Phillips curve is

$$\pi_t = \frac{\alpha \lambda}{1 - \lambda} y_t + \lambda \sum_{j=0}^{\infty} (1 - \lambda)^j E_{t-1-j} (\pi_t + \alpha \Delta y_t), \quad (18)$$

where $\Delta y_t = y_t - y_{t-1}$.

Interpretation. The critical difference is the timing of expectations. Inflation depends on *past expectations of current conditions*. This is why inflation adjusts sluggishly even with rational expectations.

3.4 Why strategic complementarity matters

Using aggregate demand, the desired price can be written as

$$p_t^* = (1 - \alpha)p_t + \alpha m_t. \quad (19)$$

If $\alpha = 1$, the desired price only depends on aggregate demand, so informed firms move immediately when policy changes. Inflation then adjusts quickly.

If $\alpha < 1$, firms also care about the overall price level. Even informed firms therefore respond cautiously until many other firms have learned the new information and adjusted their plans.

Interpretation. Real rigidity or strategic complementarity amplifies the inertia generated by slow information updating.

3.5 The McCallum critique and the natural-rate property

McCallum criticized the standard sticky-price model because a permanent fall in inflation can keep output permanently high, violating a strict natural-rate view.

The sticky-information model avoids this. In the absence of surprises,

$$p_t = E_{t-j} p_t \quad (20)$$

for updating lags, which implies

$$y_t = 0. \quad (21)$$

Interpretation. Slow information diffusion can generate short-run nonneutrality without implying that policy can permanently raise real activity.

3.6 Experiment 1: a permanent drop in the level of aggregate demand

The first experiment is an unexpected permanent 10% fall in the level of m_t at time 0.

Main results:

- All three models produce a recession.
- In the sticky-price model, inflation responds immediately.
- In the sticky-information model, inflation responds gradually, with the maximum effect after seven quarters.
- Output is close to baseline again after about sixteen quarters.

Interpretation. Inflation inertia emerges in the sticky-information model without assuming backward-looking expectations. This is the first clear gain relative to the standard sticky-price model.

3.7 Experiment 2: an unanticipated disinflation

The second experiment changes the *growth rate* of aggregate demand. Money grows at 10% per year until time 0; from time 0 onward, it is constant.

Main results:

- In the sticky-price model, inflation falls immediately and output does not move: disinflation is costless.
- In the sticky-information model, inflation falls only gradually.
- With constant money and slowly adjusting prices, real balances fall little at first, so the economy experiences a recession.
- The recession bottoms out about six quarters after the policy change and is almost gone by around twenty quarters.

Interpretation. The sticky-information model delivers contractionary disinflations, which is much closer to the historical evidence than the sticky-price benchmark.

3.8 Experiment 3: an anticipated and credible disinflation

The third experiment is the same disinflation as experiment 2, but announced eight quarters in advance.

Main results:

- In the backward-looking model, nothing changes before implementation because expectations are adaptive.
- In the sticky-price model, the announced disinflation causes a boom because inflation falls in advance, raising real balances before money growth slows.

- In the sticky-information model, there is no boom. Output and inflation do not move before the policy starts, but the recession is smaller once the policy begins.
- In the calibration, a disinflation announced eight quarters ahead has an output cost only about one-fifth as large as the surprise disinflation.

Interpretation. The sticky-information model combines two desirable features: disinflations are recessionary, but expectations and credibility still matter.

3.9 Realistic monetary shocks

The paper then studies a more realistic process,

$$\Delta m_t = 0.5\Delta m_{t-1} + \varepsilon_t. \quad (22)$$

A one-standard-deviation contractionary shock is then fed through all three models.

Main results:

- Output is hump-shaped in all three models because the AR(1) process makes demand keep moving after the initial shock.
- The key difference is inflation: in the sticky-price model, inflation reacts fastest on impact.
- In the sticky-information model, inflation reaches its maximum response only after seven quarters.

Interpretation. This delayed response is the paper's headline success. It matches central-bank intuition and empirical VAR evidence far better than the standard sticky-price model.

3.10 Inflation persistence

Using the same AR(1) money-growth process, the paper computes the first eight autocorrelations of inflation under each model.

Main result:

- All three models generate high inflation autocorrelation once money growth itself is serially correlated.

Interpretation. Persistence alone is not enough to distinguish models. The real discriminatory fact is the *timing* of inflation's response to monetary shocks.

3.11 The acceleration phenomenon

The paper defines the acceleration phenomenon as the positive correlation between the level of economic activity and the *change* in inflation.

In U.S. quarterly data from 1960 to 1999, the correlation between detrended output and the one-year centered change in inflation is positive:

- GDP deflator: 0.48,

- CPI: 0.38,
- core CPI: 0.46.

For the two-year inflation change, the correlations are even larger: 0.60, 0.46, and 0.51.

Predicted by the models:

- backward-looking model: nearly perfect positive correlation,
- sticky-price model: small negative correlation (about -0.13 and -0.11),
- sticky-information model: strongly positive correlation (about 0.43 and 0.40).

Interpretation. The sticky-price model gets the sign wrong because inflation falls immediately after a contractionary shock and then later starts rising while output is still weak. The sticky-information model gets the sign right because inflation keeps falling while output is depressed.

4 Identification and instruments (what makes the estimates credible)

4.1 There is no external instrument

This is a theory paper, so credibility does not come from an IV or quasi-experimental design. Instead, credibility comes from:

- transparent comparison of rival models under the same monetary experiments,
- confronting each model with facts that macroeconomists broadly accept,
- and checking whether the sticky-information mechanism survives when the environment becomes more realistic.

Interpretation. The question is not “what is the reduced-form treatment effect?” It is “which structural mechanism generates the right dynamic patterns?”

4.2 What identifies the baseline comparison

The authors compare the sticky-price and sticky-information models holding fixed:

- the desired price equation,
- aggregate-demand closure,
- and the key parameters α and λ .

This makes the difference in results come from one place: whether rigidity is located in price adjustment or in information updating.

Interpretation. The paper’s comparison is clean because both models are deliberately made as parallel as possible.

4.3 Why the sticky-information model looks more credible than the sticky-price model

The model does better on three central facts:

- disinflations are contractionary rather than expansionary,
- monetary shocks affect inflation with a long lag,
- and output is positively related to the change in inflation.

The sticky-price model fails on all three margins in the baseline calibration.

Interpretation. The paper's core claim is comparative: sticky information is not proven true, but it systematically dominates the standard sticky-price alternative on the facts the profession cares about.

4.4 Evidence on regime changes and credibility

The authors argue that the sticky-information model differs from a purely backward-looking model in two ways.

First, regime changes matter. Ball's evidence shows that the reduced-form Phillips curve appears to differ between the gold-standard era and the postwar era. That is consistent with rational expectations plus slow information updating, but not with a fixed adaptive-expectations rule.

Second, credibility matters. Boschen and Weise find that higher credibility lowers the output cost of disinflation, which is consistent with sticky information but not with the backward-looking benchmark.

Interpretation. Sticky information preserves the policy relevance of announcements and credibility while still generating inertia.

4.5 Evidence on slow dissemination of information

The paper points to several pieces of suggestive evidence.

- Carroll compares inflation expectations of professional forecasters and the general public, finding that public expectations adjust to professional expectations with a lag.
- The estimated updating parameter is remarkably close to the paper's calibration of $\lambda = 0.25$.
- Zbaracki et al. show that the main costs associated with price changes are not literal menu costs, but managerial and customer costs tied to information gathering, decision-making, negotiation, and communication.

Interpretation. These findings do not prove the model, but they point toward sluggish information processing as a plausible source of nominal rigidity.

4.6 Microfoundations and bounded rationality

The paper treats microfoundations as open, not finished. It discusses two broad possibilities:

- **limited-information channels** in the spirit of Sims and Woodford, where agents receive noisy signals about monetary policy;
- **costly thinking**, where people could in principle observe policy but find it costly to work out what it implies and thus update plans only occasionally.

The authors lean toward the second interpretation: the binding friction may not be hearing what the central bank did, but processing what it means.

Interpretation. The paper is an early bounded-rationality Phillips curve: the rigidity lies in mental updating rather than in literal price stickers.

4.7 What the paper can and cannot distinguish

What the paper does well:

- cleanly separates sticky prices from sticky information,
- shows why the timing of inflation matters more than persistence alone,
- and clarifies the role of credibility and announcements.

What it does not do:

- it does not include supply shocks,
- it keeps the information arrival rate λ constant over time,
- it uses a deliberately simple quantity-theory aggregate-demand block,
- and it does not derive the information friction from fully explicit optimizing foundations.

Interpretation. The paper is best read as a disciplined replacement benchmark, not as the last word on bounded rationality or monetary transmission.

4.8 Relation to the literature

The sticky-information model sits between several traditions:

- **Calvo/Taylor new Keynesian pricing:** keeps monopolistic competition and nominal rigidity but relocates the friction from prices to information.
- **Fischer-style contracts:** resembles their dynamics because past expectations matter for current inflation.
- **Backward-looking Phillips curves:** matches their recessionary disinflations and inertial inflation, but keeps rational expectations and credibility effects.
- **Lucas/Sims/Woodford imperfect-information models:** shares the focus on information frictions, though with different assumptions about how information arrives.

5 Tables and figures

5.1 Figure I: a permanent fall in the level of aggregate demand

Read.

- Output falls in all three models after an unexpected 10% drop in the level of aggregate demand.
- The sticky-price model and sticky-information model both yield monotonic output recoveries; the backward-looking model is more oscillatory.
- The real difference is in inflation: the sticky-price model responds almost immediately, while the sticky-information model adjusts much more slowly, with the largest inflation effect after about seven quarters.

Economic message. The figure isolates the paper's basic mechanism: delayed inflation adjustment can arise from sluggish information updating rather than adaptive expectations.

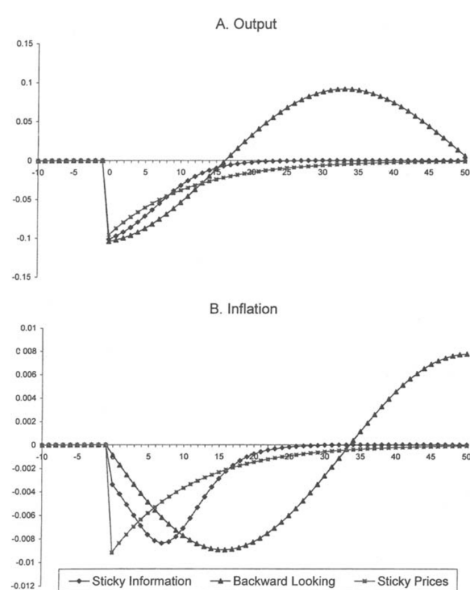


FIGURE I
Dynamic Paths after a 10 Percent Fall in the Level of Aggregate Demand at Time 0

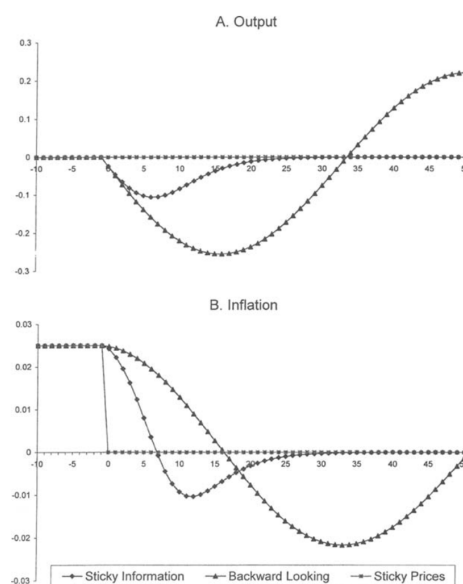


FIGURE II
Dynamic Paths Given an Unanticipated Fall in the Growth Rate of Aggregate Demand at Date 0

5.2 Figure II: an unanticipated disinflation

Read.

- In the sticky-price model, inflation jumps down immediately and output barely moves.
- In the sticky-information model, inflation declines slowly and output falls into recession, with the trough around six quarters after the policy change.
- The backward-looking model also produces a recession, but by construction through adaptive expectations.

Economic message. This figure is the direct rebuttal to the costless-disinflation logic embedded in the standard sticky-price model.

5.3 Figure III: an anticipated disinflation

Read.

- The sticky-price model predicts a boom before implementation because inflation falls in anticipation.
- The sticky-information model predicts no pre-implementation boom, but the recession after implementation is milder than under a surprise disinflation.
- The backward-looking model is unchanged because announcements do not matter under adaptive expectations.

Economic message. Sticky information combines inflation inertia with rational expectations: credibility matters, but it does not generate counterfactual expansionary disinflations.

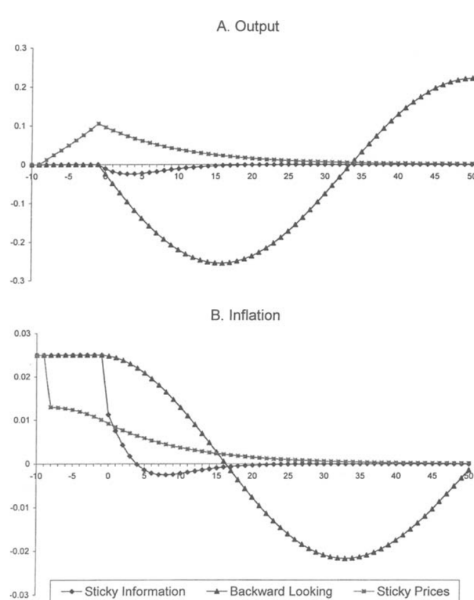


FIGURE III
Dynamic Paths Given an Announcement at Date -8 of a Fall in the Growth Rate of Aggregate Demand at Date 0

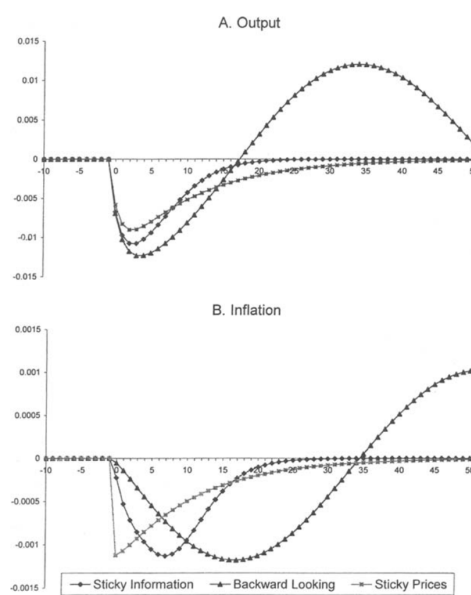


FIGURE IV
Dynamic Paths after a Negative One Standard Deviation (-0.007) Shock to the AR(1) Aggregate Demand

5.4 Figure IV: impulse responses to a realistic monetary shock

Read.

- Output is hump-shaped in all three models because the monetary shock itself builds over time.
- The sticky-price model again predicts that inflation reacts immediately.
- The sticky-information model delays the peak inflation response to about seven quarters.

Economic message. The figure shows the model's most policy-relevant success: it can reproduce the long lag between monetary policy and inflation emphasized by central bankers and VAR evidence.

5.5 Table I: inflation autocorrelations

Read.

- All three models generate high autocorrelations of inflation when money growth follows the realistic AR(1) process.
- For example, the sticky-information model generates first-order autocorrelation 0.99, while the GDP-deflator series in the data has 0.89.
- The sticky-price model is also persistent once money growth is persistent.

Economic message. Persistence by itself is not the decisive test. The sticky-price model's real problem is the timing of inflation's response, not the unconditional level of autocorrelation.

TABLE I
AUTOCORRELATIONS FOR INFLATION: PREDICTED AND ACTUAL

	Sticky- information model	Sticky- price model	Backward- looking model	Actual: GDP deflator	Actual: CPI	Actual: core CPI
1	0.99	0.92	0.99	0.89	0.76	0.76
2	0.95	0.85	0.98	0.83	0.72	0.71
3	0.89	0.78	0.96	0.81	0.73	0.69
4	0.82	0.71	0.94	0.78	0.62	0.59
5	0.74	0.65	0.90	0.71	0.57	0.55
6	0.66	0.59	0.86	0.65	0.51	0.54
7	0.57	0.54	0.81	0.61	0.44	0.46
8	0.48	0.50	0.75	0.58	0.33	0.38

The first three columns of this table show the autocorrelations of inflation predicted by three models. These calculations assume that money growth follows the process $\Delta m_t = 0.5 \Delta m_{t-1} + \epsilon_t$. The model parameters are set to $\alpha = .1$ and $\lambda = .25$. The last three columns show the actual autocorrelations of quarterly inflation rates.

TABLE II
THE ACCELERATION PHENOMENON

	$\text{corr}(y_t, \pi_{t-2} - \pi_{t-2})$	$\text{corr}(y_t, \pi_{t+4} - \pi_{t-4})$
A. Actual		
GDP deflator	.48	.60
CPI	.38	.46
Core CPI	.46	.51
B. Predicted		
Backward-looking model	.99	.99
Sticky-price model	-.13	-.11
Sticky-information model	.43	.40

Panel A shows the correlation between output and the change in inflation in U. S. quarterly data from 1960 to 1999. The variable y is measured as log real GDP detrended with the Hodrick-Prescott filter. Panel B shows the correlation between output and the change in inflation predicted by three models. These correlations assume that money growth follows the process: $\Delta m_t = 0.5 \Delta m_{t-1} + \epsilon_t$. The model parameters are set to $\alpha = .1$ and $\lambda = .25$.

5.6 Table II: the acceleration phenomenon

Read.

- In U.S. data, detrended output is positively correlated with subsequent changes in inflation.
- The sticky-information model predicts positive correlations close to the data: about 0.43 and 0.40.
- The sticky-price model predicts the wrong sign: roughly -0.13 and -0.11 .
- The backward-looking model predicts very high positive correlations by construction.

Economic message. This table is one of the paper's strongest pieces of evidence: the sticky-information model matches the sign and rough magnitude of a classic reduced-form Phillips-curve fact.

5.7 Table III: sensitivity analysis for the acceleration result

Read.

- Across alternative values of α and λ , the sticky-price model typically continues to imply small negative correlations between output and changes in inflation.
- Across the same range, the sticky-information model usually implies positive correlations, though the magnitude varies with strategic complementarity and the updating rate.

TABLE III
THE ACCELERATION PHENOMENON: SENSITIVITY ANALYSIS

A. Sticky-price model				
	$\alpha = .05$	$\alpha = .1$	$\alpha = .5$	$\alpha = 1.0$
$\lambda = .1$	-0.08	-0.09	-0.12	-0.13
$\lambda = .25$	-0.12	-0.13	-0.15	-0.15
$\lambda = .5$	-0.15	-0.15	-0.13	-0.11
B. Sticky-information model				
	$\alpha = .05$	$\alpha = .1$	$\alpha = .5$	$\alpha = 1.0$
$\lambda = .1$	0.49	0.39	0.05	-0.04
$\lambda = .25$	0.51	0.43	0.12	0.02
$\lambda = .5$	0.52	0.44	0.21	0.13

This table shows the correlation between output y_t and the change in inflation $\pi_{t+2} - \pi_{t-2}$. These correlations assume money growth follows the process: $\Delta m_t = 0.5 \Delta m_{t-1} + \epsilon_t$.

Economic message. The acceleration result is not a knife-edge feature of one calibration. It is robust to a broad range of plausible parameter values.

6 Short summary

- The paper replaces sticky prices with sticky information as the source of nominal rigidity.
- The resulting sticky-information Phillips curve depends on current output and on lagged information sets about current inflation and output growth.
- Compared with the standard sticky-price model, it gives much more realistic monetary dynamics: disinflations are contractionary, inflation responds to monetary shocks with a delay, and output is positively related to changes in inflation.
- Relative to a backward-looking Phillips curve, it keeps rational expectations, so announcements and credibility matter.
- The paper does not claim to have a fully developed microfoundation, but it offers bounded rationality and costly information processing as promising foundations.
- The broad lesson is that inflation inertia may come from slow updating of plans rather than from literal inability to change prices.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

The paper establishes that a sticky-information model can outperform the standard sticky-price New Keynesian Phillips curve on several core monetary-policy facts. In particular, it generates recessionary disinflations, delayed inflation responses to monetary shocks, and a positive relation between output and changes in inflation.

7.2 Economic intuition

The key intuition is simple: firms may be perfectly able to post a price today, yet still base that price on stale information and old plans. If only a subset of firms processes new information each period,

inflation responds gradually even with rational expectations. Strategic complementarities then reinforce this gradual adjustment, because informed firms also care about what the still-uninformed firms are charging.

7.3 Takeaway

Mankiw and Reis offer a powerful reorientation of New Keynesian nominal rigidity. The central friction may be less about menu costs and more about the slow diffusion, interpretation, and use of information. That shift in perspective goes a long way toward reconciling Phillips-curve theory with how monetary policy seems to work in practice.